JAVA PROGRAMMING

LAB Exercise Unit 4

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Exercise 1:

```
import java.util.Scanner;

public class 2la {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();
        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();
        System.out.print("Enter third number: ");
        int num3 = sc.nextInt();

        int largest;

        if (num1 >= num2 && num1 >= num3) {
            largest = num1;
        } else if (num2 >= num1 && num2 >= num3) {
            largest = num2;
        } else {
            largest = num3;
        }

        System.out.println("The largest number is: " + largest);
        sc.close();
    }
```

}

```
PROBLEMS 21   OUTPUT   DEBUG CONSOLE   TERMINAL   POR

PS E:\Tina DeskTop\Java>  & 'C:\Program Files\Java\jd
iew' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
a_62d56fb3\bin' 'LargestNumber'
Enter first number: 2
Enter second number: 5
Enter third number: 6
The largest number is: 6
PS E:\Tina DeskTop\Java>
```

Exercise 2:

```
import java.util.Scanner;


public class 2lb {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();
        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();
        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);


        double result;


        switch (operator) {
            case '+':
```

```
            result = num1 + num2;
            System.out.println(num1 + " + " + num2 + " = " + result);
            break;
        case '-':
            result = num1 - num2;
            System.out.println(num1 + " - " + num2 + " = " + result);
            break;
        case '*':
            result = num1 * num2;
            System.out.println(num1 + " * " + num2 + " = " + result);
            break;
        case '/':
            if (num2 != 0) {
                result = num1 / num2;
                System.out.println(num1 + " / " + num2 + " = " + result);
            } else {
                System.out.println("Error: Division by zero!");
            }
            break;
        default:
            System.out.println("Error: Invalid operator!");
    }

    sc.close();
  }
}
```

```
PROBLEMS 21    OUTPUT    DEBUG CONSOLE    TERMINA

PS E:\Tina DeskTop\Java>  & 'C:\Program Files\
ing\Code\User\workspaceStorage\637b3b355c2760b
Enter first number: 23
Enter second number: 76
Enter operator (+, -, *, /): +
23.0 + 76.0 = 99.0
PS E:\Tina DeskTop\Java>
```

Exercise 3:

```
import java.util.Scanner;


public class 2lc {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();

        // Pattern 1: Right-angled triangle
        System.out.println("\nPattern 1: Right-angled triangle");
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }

        // Pattern 2: Inverted right-angled triangle
        System.out.println("\nPattern 2: Inverted triangle");
```

```
        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }

        // Pattern 3: Pyramid
        System.out.println("\nPattern 3: Pyramid");
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2 * i - 1; k++) {
                System.out.print("*");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

```
a_62d56fb3\bin' 'Patterns'
Enter number of rows: 9

Pattern 1: Right-angled triangle
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * * *
* * * * * * * * *

Pattern 2: Inverted triangle
* * * * * * * * *
* * * * * * * *
* * * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*

Pattern 3: Pyramid
        *
       ***
      *****
     *******
    *********
   ***********
  *************
 ***************
*****************
PS E:\Tina DeskTop\Java>
```

⊗ 20 ⚠ 0 ⓘ 1 Java: Warning

Exercise 4:

```
import java.util.Scanner;


public class 2le {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
        int number = sc.nextInt();
        int original = number;
        int reversed = 0;

        while (number != 0) {
            int digit = number % 10;
            reversed = reversed * 10 + digit;
            number = number / 10;
        }

        System.out.println("Original number: " + original);
        System.out.println("Reversed number: " + reversed);

        sc.close();
    }
}
```

```
PS E:\Tina DeskTop\Java>  & 'C:\Program Files\J
iew' '-XX:+ShowCodeDetailsInExceptionMessages'
a_62d56fb3\bin' 'ReverseNumber'
Enter a number: 3459
Original number: 3459
Reversed number: 9543
PS E:\Tina DeskTop\Java>
```

Exercise 5:

```
import java.util.Scanner;
```

```
public class PrimeChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        if (isPrime(number)) {
            System.out.println(number + " is a prime number.");
        } else {
            System.out.println(number + " is not a prime number.");
        }

        sc.close();
    }

    public static boolean isPrime(int n) {
        if (n <= 1) {
            return false;
        }
        if (n == 2) {
            return true;
        }
        if (n % 2 == 0) {
            return false;
        }
```

```
        for (int i = 3; i <= Math.sqrt(n); i += 2) {
            if (n % i == 0) {
                return false;
            }
        }

        return true;
    }
}
```

```
PS E:\Tina DeskTop\Java>  & 'C:\Program Files
ing\Code\User\workspaceStorage\637b3b355c2760
Enter a number: 56002
56002 is not a prime number.
PS E:\Tina DeskTop\Java>
```